# Lecture 4A: Linear Regression I

## From likelihood to least squares, projection, and uniqueness

**Instructor: Prof. Udit Bhatia**

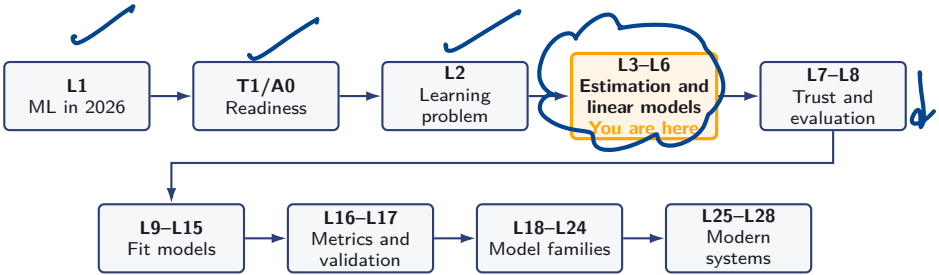
Civil Engineering; Computer Science and Engineering

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# Course Roadmap: You Are Here




Linear Regression I: conditional model → fitting objective → parameter estimate → geometric projection.

# A Question From The Previous Class

**Question**

Why can a modern language model write complex software, yet fail on a simple-looking number sequence?

**"Complex" is not one mathematical quantity.**

- Familiar code has strong patterns in training data.
- A short sequence can support many incompatible rules.
- The prompt may not identify which rule generated it.

**The model still needs a criterion.**

- What family of explanations is allowed?
- What counts as a costly error?
- Which parameter value is preferred?

**Insight**

Today we make those three choices explicit: model family, loss, and estimator.

# Vocabulary We Will Reuse Today

| Idea | Meaning in this lecture |
|---|---|
| Conditional probability | A distribution for an output after an input is known: $p(y \mid x)$. |
| Non-stationarity | The data-generating mechanism may change with time or context. |
| Weights / parameters | Unknown numbers $\theta$ that determine a member of a model family. |
| Optimization | A rule for choosing $\theta$, usually by minimizing an objective. |
| Convexity | A property of the objective as a function of a specified variable. |

**Warning**

Never say "the problem is convex" without naming the variable: convex in $x$, in $\theta$, or in something else?

# Learning Objectives

By the end of the lecture, you should be able to:

1. distinguish a parameter, estimator, estimate, fitted model, and prediction;
2. derive squared-error fitting from a Gaussian conditional model;
3. explain why another noise model can produce another loss;
4. derive the normal equations and the projection matrix;
5. prove the residual is orthogonal to the design space;
6. test whether least squares is convex in $\theta$, and state when its minimizer is unique.



# One Dataset For The Entire Lecture

Suppose $x_i$ is batch size in thousands of rows and $y_i$ is response time in milliseconds.

| $i$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $x_i$ | 0 | 1 | 2 | 3 |
| $y_i$ | 2 | 5 | 6 | 9 |

The observed dataset is

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}, \qquad n = 4.$$

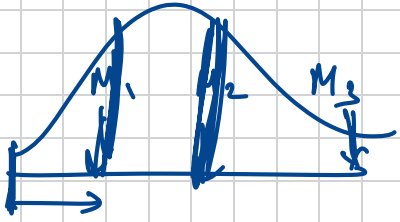
**Discriminative regression**

We model the conditional behaviour of the output given the input: $p(y \mid x)$, or a summary such as $f(x) \approx \mathbb{E}[Y \mid X = x]$.

$E(y|x) = ?$ $p(x)$: pdf




=




$M_A =$

$E(x^2) = \sum x^2 p(x)$

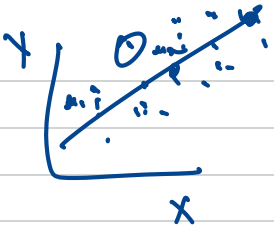
$\vdots$

$E(x^n) =$

$\leftrightarrow E(x) = \sum_x x\, p(x)$

mean

$Y_i \propto X_i$




$Y_i = \omega X_i + \varepsilon_i$ ← pdf ← GWN

$\omega$ ? $X_i$ ↙

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$Y = \omega X + \varepsilon \quad (\varepsilon \sim N(0, \sigma^2))$$

$$E(Y) = E(\omega X) + E(\varepsilon)$$

$$\downarrow$$

$$E(Y|X) = \omega X + E(N(0, \sigma^2)) \quad (E(N(0,\sigma^2)) = 0)$$

$$=$$

$$E(Y|X) = \omega X$$

$\hat{y} = wx + \epsilon$

$n = 4$

$$p(Y|X) = \mathcal{N}(wx, \sigma^2)$$

$$p(\hat{Y}|X) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(Y - \hat{w}x)^2}{2\sigma^2}}$$

$$\hookrightarrow L = \prod_{i=1}^{n=4} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \hat{w}x_i)^2}{2\sigma^2}}$$

$\omega \equiv \theta \leftarrow \hat{y} = \theta x$ or $\omega x$

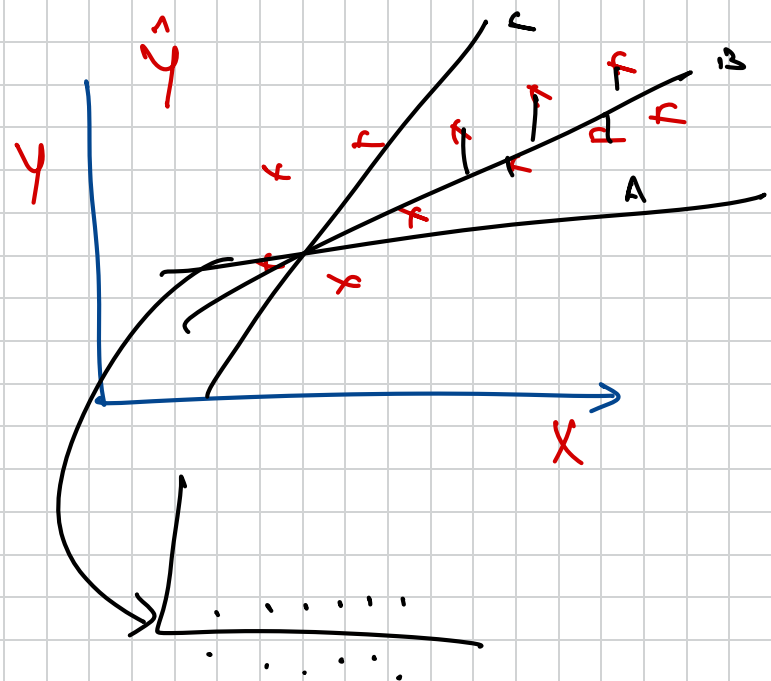
max

$\log(L(\theta)) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y-\hat{y})^2$

$-\log(L(\theta)) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y-\hat{y})^2$

min.

min. $\sum_{i=1}^{n}(y-\hat{y})^2$


y
x
A
B

# From A Fixed Mean To A Rule That Depends On $x$

Previously, a Gaussian centre was one unknown scalar:

$$Y_i \sim \mathcal{N}(\mu, \sigma^2).$$

Now the centre changes with the input:

$$Y_i \mid x_i \sim \mathcal{N}\left(f_{\boldsymbol{\theta}}(x_i), \sigma^2\right), \qquad f_{\boldsymbol{\theta}}(x) = \theta_0 + \theta_1 x.$$

**Parameter vector**

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} \in \mathbb{R}^2$$

**Interpretation**

- $\theta_0$: fixed overhead;
- $\theta_1$: additional time per thousand rows.

# Five Objects That Must Not Be Confused

| Object | Notation | Meaning |
|---|---|---|
| Parameter | $\theta$ | Unknown population quantity inside the model family. |
| Estimator | $\widehat{\theta}(\mathcal{D})$ | Rule that maps a dataset to a parameter value. |
| Estimate | $\widehat{\theta}$ | Numerical output after applying the rule to this dataset. |
| Fitted model | $f_{\widehat{\theta}}$ | Function obtained after the estimate is inserted. |
| Prediction | $y_*$ | Output $f_{\widehat{\theta}}(x_*)$ for a specified input. |

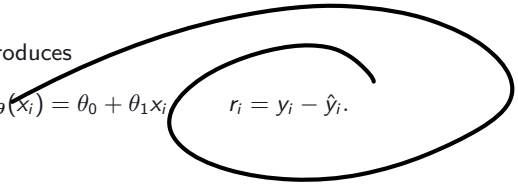
**Insight**

The optimizer does not "learn a dataset." It applies an estimator to the dataset and returns parameter estimates.

# A Residual Measures One Miss

For row $i$, a candidate parameter vector produces

$$\hat{y}_i = f_{\theta}(x_i) = \theta_0 + \theta_1 x_i \qquad r_i = y_i - \hat{y}_i.$$



Try the candidate line $f_{\theta}(x) = 1 + 1.5x$:

| $x_i$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y_i$ | 2 | 5 | 6 | 9 |
| $\hat{y}_i$ | 1 | 2.5 | 4 | 5.5 |
| $r_i = y_i - \hat{y}_i$ | 1 | 2.5 | 2 | 3.5 |

Residuals retain sign. A **loss** converts a residual into a nonnegative cost.

# The Data Do Not Choose The Loss

| Loss | Formula | What it emphasizes |
|---|---|---|
| Squared | $r^2$ | Large errors receive rapidly increasing cost. |
| Absolute | $\lvert r \rvert$ | Every additional unit of error costs the same. |
| Huber | quadratic then linear | Smooth near zero; less sensitive in the tails. |
| Asymmetric | different cost by sign | Under-prediction and over-prediction need not be equivalent. |

**Question**

Is the loss a mathematical convenience, a probabilistic assumption, or a description of real decision cost?

It can be any of the three. A professional analysis says which interpretation is being used.

# From One Loss To An Objective

For a parameter vector $\boldsymbol{\theta}$, define the average training loss

$$\widehat{R}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f_{\boldsymbol{\theta}}(x_i)) \,.$$

The quantity we ultimately care about is the population risk

$$R(\boldsymbol{\theta}) = \mathbb{E}_{(X,Y)\sim P} \left[ \ell(Y, f_{\boldsymbol{\theta}}(X)) \right].$$

**Available**

$$\widehat{R}_n(\boldsymbol{\theta})$$

computed from observed data.

**Desired**

$$R(\boldsymbol{\theta})$$

performance on the generating population.

# Parameter Fitting Is An Optimization Rule

Empirical risk minimization chooses

$$\widehat{\boldsymbol{\theta}} \in \arg\min_{\boldsymbol{\theta}\in\Theta} \widehat{R}_n(\boldsymbol{\theta}).$$




**Warning**

Model, objective, optimizer, and evaluation metric are different objects. They may be chosen together, but they are not interchangeable.

# Maximum Likelihood Is Also An Optimization Rule

Let $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$ collect the observed inputs. Assume conditionally independent observations:

$$p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) = \prod_{i=1}^{n} p(y_i \mid x_i, \boldsymbol{\theta}).$$

Maximum likelihood chooses

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MLE}} \in \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}).$$

Because $-\log(\cdot)$ is strictly decreasing,

$$\arg\max_{\boldsymbol{\theta}} p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \left[-\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta})\right].$$

**Insight**

Negative log likelihood is a loss-generating mechanism. The assumed distribution determines its mathematical shape.

# One Gaussian Observation Contains A Square

Assume

$$Y_i \mid x_i \sim \mathcal{N}(\theta_0 + \theta_1 x_i, \sigma^2).$$

Then

$$p(y_i \mid x_i, \boldsymbol{\theta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \theta_0 - \theta_1 x_i)^2}{2\sigma^2}\right].$$

The residual appears squared in the exponent:

$$r_i(\boldsymbol{\theta}) = y_i - \theta_0 - \theta_1 x_i.$$

**Question**

What happens to the product of $n$ Gaussian densities after taking a logarithm?

# From A Product To A Sum

**Start with**

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{r_i(\boldsymbol{\theta})^2}{2\sigma^2}\right].$$

Take the logarithm and simplify every step.

**Use** $\log(\prod_i a_i) = \sum_i \log a_i$.

Let $p_i(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{r_i(\theta)^2}{2\sigma^2}\right]$.

$$\begin{aligned}
\log L(\theta) &= \log\left(\prod_{i=1}^{n} p_i(\theta)\right) \\
&= \sum_{i=1}^{n} \log p_i(\theta) \\
&= \sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi\sigma^2) \right. \\
&\qquad \left. - \frac{r_i(\theta)^2}{2\sigma^2}\right] \\
&= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n} r_i(\theta)^2.
\end{aligned}$$

# The Gaussian MLE Becomes Least Squares

Starting from the log likelihood, identify which terms depend on $\boldsymbol{\theta}$.
Then convert maximization into an equivalent minimization.

**Turn maximization into minimization:**

$$\arg\max_{\theta} \log L(\theta) = \arg\min_{\theta}\{-\log L(\theta)\}.$$

For fixed $\sigma^2$, terms independent of $\theta$ form $C$:

$$-\log L(\theta) = C + \frac{1}{2\sigma^2}\sum_{i=1}^{n}[y_i - f_\theta(x_i)]^2.$$

Drop $C$ and the positive scale; the minimizer is unchanged:

$$\begin{aligned}\widehat{\theta}_{\mathrm{MLE}} &\in \arg\min_{\theta}\sum_{i=1}^{n}[y_i - f_\theta(x_i)]^2 \\ &= \arg\min_{\theta}\mathrm{SSE}(\theta).\end{aligned}$$

# SSE, MSE, And RMSE Fit The Same Parameters

Define

$$\mathrm{SSE}(\boldsymbol{\theta}) = \sum_{i=1}^{n} r_i(\boldsymbol{\theta})^2,$$
$$\mathrm{MSE}(\boldsymbol{\theta}) = \frac{1}{n}\mathrm{SSE}(\boldsymbol{\theta}),$$
$$\mathrm{RMSE}(\boldsymbol{\theta}) = \sqrt{\mathrm{MSE}(\boldsymbol{\theta})}.$$

Since multiplication by $(1/n>0)$ and the square root are strictly increasing,

$$\arg\min_{\boldsymbol{\theta}} \mathrm{SSE} = \arg\min_{\boldsymbol{\theta}} \mathrm{MSE} = \arg\min_{\boldsymbol{\theta}} \mathrm{RMSE}.$$

**Warning**

The fitted parameters agree, but the reported values do not: RMSE has the same physical units as $y$; MSE does not.

# Noise And Residual Are Not The Same Object

The probabilistic model introduces an unobserved random error:

$$Y_i = f_\theta(x_i) + \varepsilon_i.$$

| Object | Notation | Meaning |
|---|---|---|
| Noise / error | $\varepsilon_i$ | Random quantity in the data-generating model; it is not directly observed. |
| Residual | $r_i = y_i - f_{\hat{\theta}}(x_i)$ | Number calculated after fitting; it depends on the observed data and fitted parameters. |

A distributional assumption is made about $\varepsilon_i$, not about a residual that has already been calculated.

**Insight**

Residual plots are evidence used to assess whether the assumed noise model is reasonable.

# The Loss Can Change The Fitted Answer

Fit one constant $a$ to five observations:

$$2,\ 3,\ 4,\ 5,\ 20.$$

**Squared loss**

$$J_2(a) = \sum_{i=1}^{5} (y_i - a)^2.$$

**Absolute loss**

$$J_1(a) = \sum_{i=1}^{5} |y_i - a|.$$

Find each minimizer. Which one is pulled farther by 20?

# Gaussian Noise Produces Squared Loss

Assume independent Gaussian errors:

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2), \qquad p(\varepsilon_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_i^2}{2\sigma^2}\right).$$

Substitute $\varepsilon_i = y_i - f_{\boldsymbol{\theta}}(x_i)$, take the negative logarithm, and sum:

$$-\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^{n} [y_i - f_{\boldsymbol{\theta}}(x_i)]^2 .$$

For fixed $\sigma^2$, the first term and positive scale do not affect the minimizer:

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MLE}} \in \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} [y_i - f_{\boldsymbol{\theta}}(x_i)]^2 .$$

# Laplace Noise Produces Absolute Loss

The Laplace distribution has a sharper centre and heavier tails than a Gaussian:

$$\varepsilon_i \sim \mathrm{Laplace}(0, b), \qquad p(\varepsilon_i) = \frac{1}{2b} \exp\left(-\frac{|\varepsilon_i|}{b}\right), \qquad b > 0.$$

Again substitute $\varepsilon_i = y_i - f_\theta(x_i)$:

$$-\log p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{\theta}) = n \log(2b) + \frac{1}{b} \sum_{i=1}^{n} |y_i - f_\theta(x_i)| .$$

Therefore

$$\widehat{\boldsymbol{\theta}}_{\mathrm{MLE}} \in \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} |y_i - f_\theta(x_i)| .$$

**Insight**

Laplace noise is an assumption that may suit sharply centred, heavy-tailed errors. It does not automatically arise from every dataset.

# So Why Use Squared Error Rather Than Absolute Error?

| | Squared loss | Absolute loss | Huber loss |
|---|---|---|---|
| Error model | Gaussian | Laplace | Gaussian-like centre; Laplace-like tails |
| Population target | conditional mean | conditional median | compromise controlled by $\delta$ |
| Large residuals | strongly amplified | grow linearly | quadratic, then linear |
| Optimization | smooth | corner at zero | smooth transition |

Huber loss makes the compromise explicit:

$$\ell_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\left(|r| - \frac{1}{2}\delta\right), & |r| > \delta. \end{cases}$$

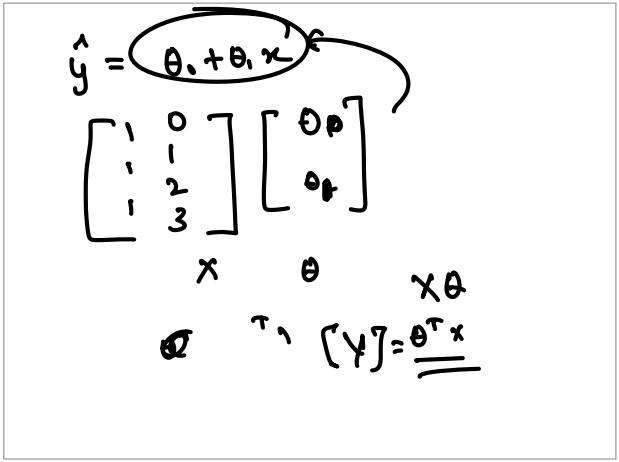
**There is no universal winner.** Choose the loss from the target quantity, plausible error process, real decision cost, and optimization behavior.

# Where Does The Design Matrix $X$ Come From?

For observation $i$,

$$\widehat{y}_i = \theta_0 + \theta_1 x_i = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}.$$

The entry 1 multiplies the intercept; $x_i$ multiplies the slope. Therefore row $i$ of $\boldsymbol{X}$ is $\begin{bmatrix} 1 & x_i \end{bmatrix}$.
For $x_i \in \{0, 1, 2, 3\}$, stack the four rows and verify every dimension.



# The Objective Is A Function Of The Parameters

The previous slide gave

$$\widehat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\theta}, \qquad \boldsymbol{r} = \boldsymbol{y} - \widehat{\boldsymbol{y}}.$$

Once the dataset is observed, $\boldsymbol{X}$ and $\boldsymbol{y}$ are fixed. The variable is $\boldsymbol{\theta}$:

$$J(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\|_2^2.$$

**Input space:** $x \in \mathbb{R}$
The fitted rule is a line in $x$.

**Parameter space:** $\boldsymbol{\theta} \in \mathbb{R}^2$
The objective is a surface over $(\theta_0, \theta_1)$.

**Warning**

Always name the variable: this objective is convex in $\boldsymbol{\theta}$, with $\boldsymbol{X}$ and $\boldsymbol{y}$ held fixed.

# Differentiate The Objective: The Normal Equations

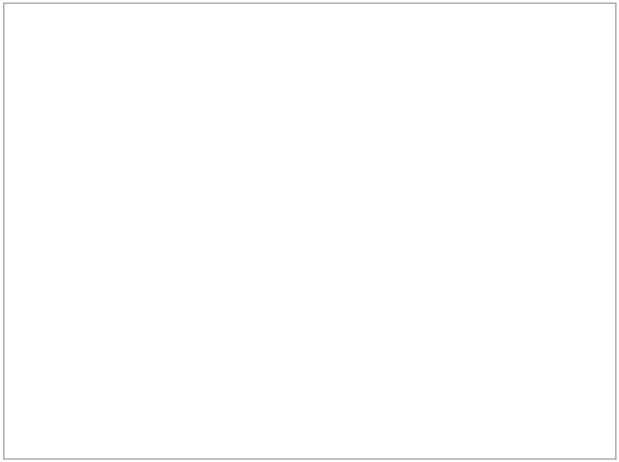
Start from

$$J(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y})^{\top}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}).$$

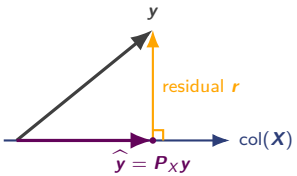
Expand, differentiate with respect to $\boldsymbol{\theta}$, and set the gradient to zero.
For the running dataset, compute $\boldsymbol{X}^{\top}\boldsymbol{X}$, $\boldsymbol{X}^{\top}\boldsymbol{y}$, and the numerical estimate.

# From Parameter Estimate To Projection Matrix

Assume the columns of $\boldsymbol{X}$ are linearly independent. Solve the normal equations for $\widehat{\boldsymbol{\theta}}$, then substitute into $\widehat{\boldsymbol{y}} = \boldsymbol{X}\widehat{\boldsymbol{\theta}}$.



# Why $P_X$ Is A Projector

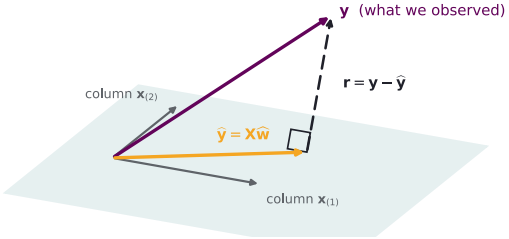



$$y \xrightarrow{P_X} \widehat{y} \xrightarrow{P_X} \widehat{y}.$$

**Idempotent:** a second projection changes nothing.
**Geometry:** $\widehat{y}$ lies in col($X$), while $r = y - \widehat{y}$ is perpendicular to it.
**ML meaning.** Features determine col($X$), the predictions the model can express. $P_X y$ is the representable part; $y - P_X y$ is not representable by this model.

# The Geometry Behind Least Squares

$\mathbf{r} \perp$ every column $\iff \mathbf{X}^\top \mathbf{r} = \mathbf{0} \iff \mathbf{X}^\top \mathbf{X}\widehat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y}$




The fitted vector must lie in the column space:

$$\widehat{\boldsymbol{y}} = \boldsymbol{X}\widehat{\boldsymbol{\theta}} \in \mathrm{col}(\boldsymbol{X}).$$

The residual is

$$\widehat{\boldsymbol{r}} = \boldsymbol{y} - \widehat{\boldsymbol{y}}.$$

From the normal equations,

$$\boldsymbol{X}^\top \widehat{\boldsymbol{r}} = \boldsymbol{X}^\top (\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\theta}}) = \mathbf{0}.$$

Thus $\widehat{\boldsymbol{r}}$ is orthogonal to every column of $\boldsymbol{X}$.

**Insight**

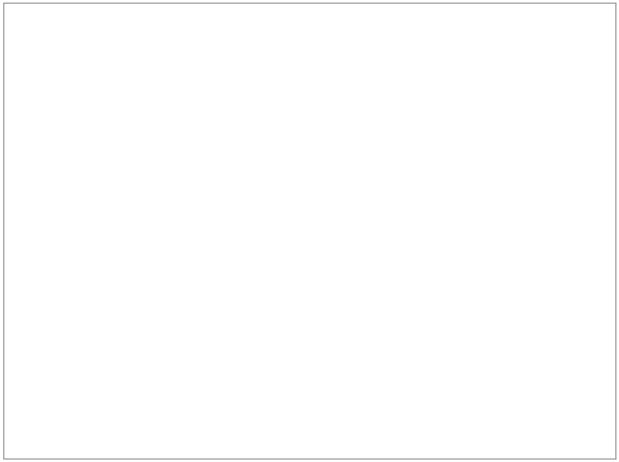
Least squares chooses the point in $\mathrm{col}(\boldsymbol{X})$ closest to $\boldsymbol{y}$ in Euclidean distance.

# Test Convexity In $\theta$

Differentiate the gradient once more:

$$\nabla_{\theta} J = \boldsymbol{X}^{\top}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}).$$

Then test the Hessian using an arbitrary vector $\boldsymbol{v}$.



# Matrix Rank Counts Independent Column Directions

For a matrix $\boldsymbol{X} \in \mathbb{R}^{n \times p}$,

$$\operatorname{rank}(\boldsymbol{X}) = \dim(\operatorname{col}(\boldsymbol{X})) .$$

It counts how many columns provide genuinely independent directions.

Compare:

$$\boldsymbol{X}_A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad \boldsymbol{X}_B = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}.$$

Find the rank of each matrix.

# A Convex Objective Can Have A Flat Valley

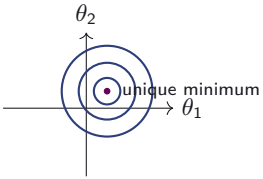
Consider two identical feature columns:

$$\boldsymbol{X} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Write $J(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\|_2^2$ using only the sum $s = \theta_1 + \theta_2$.
Find every minimizer.

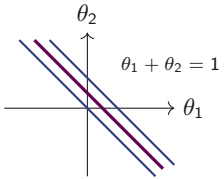
# Why Matrix Rank Controls Uniqueness

**Full rank: one bottom point**



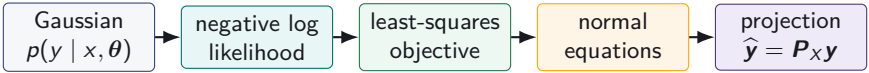

$$\text{rank}(\boldsymbol{X}) = p \iff \text{Null}(\boldsymbol{X}) = \{\mathbf{0}\}$$
$$\iff \boldsymbol{X}\boldsymbol{v} \neq \mathbf{0} \quad \forall\, \boldsymbol{v} \neq \mathbf{0}.$$
$$\boldsymbol{v}^\top \boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{v} = \|\boldsymbol{X}\boldsymbol{v}\|_2^2 > 0.$$
$$\boldsymbol{X}^\top \boldsymbol{X} \succ 0 \Longrightarrow J \text{ strictly convex}$$
$$\Longrightarrow \widehat{\boldsymbol{\theta}} \text{ unique.}$$

**Rank deficient: a flat valley**




$$\text{rank}(\boldsymbol{X}) < p \iff \exists\, \boldsymbol{v} \neq \mathbf{0} : \ \boldsymbol{X}\boldsymbol{v} = \mathbf{0}.$$
$$\boldsymbol{X}(\boldsymbol{\theta} + t\boldsymbol{v}) = \boldsymbol{X}\boldsymbol{\theta}.$$
$$J(\boldsymbol{\theta} + t\boldsymbol{v}) = J(\boldsymbol{\theta}) \Longrightarrow \text{flat valley}$$
$$\Longrightarrow \text{coefficients not unique.}$$

# Linear Regression I: The Complete Chain




The same objective also gives

$$\nabla^2 J = X^\top X \succeq 0,$$

which certifies convexity in $\theta$.

**Before The Next Class**

Explain in one sentence each: why a Gaussian assumption gives a square; why $P_X$ is a projector; and why full column rank matters.

**Next:** curves that remain linear in parameters, dimensionality, basis, eigen-directions, and why PCA is a projection chosen by the data.